EXAMINATION

Luleå University of Technology

Course: **D7024E**
Course name: **Distributed and Mobile Computing Systems**
Date: **2021-10-26**
Time: **9.00-13.00, online home exam**
Aid: **WWW (no interactive communication with others)**

Teacher on duty (complete telephone number): Olov Schelén, **0920 492237** **olov.schelen@ltu.se**	**Teacher on duty (complete telephone number):**
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Grade scale: **50% grade 3, 75% grade 4 and 90% grade 5**	Note that this grade is for the exam. The exam is supposed to be easy enough so that anyone that followed the course properly should pass it. The grade for the course (including all assessment criteria) may be different from the grade of the exam.
Total number of questions and score: **2 questions, total score 36**	
Other information: This is an individual home exam.	Note: No interactive cooperation is allowed. Information can be searched on the world wide web. Please provide references, when such information is used in you answers.

General instructions

Check that you have received all the tasks/questions. You shall formulate answers with your own words, but for parts of the answers you may cite sources on the web. In any case you must provide the references to your primary sources (e.g., URL).

After examination

Examination results are posted in Canvas within 20 workdays after the examination.

1. **Mixed question on distributed systems (12p)**

   These are multi-choice questions, with three alternatives. Please answer "true", "false" or “depends” to each of the following statements. If your answer is “depends” you must provide a short explanation to get credits. For “true” or “false”, no explanation is required, unless you want to explain what conditions are assumed. If you have identified a corner case and provide a good explanation, we can accept an answer that is different from the expected one. Each question is worth 0.5p.

   a. Distributed systems must be decentralized.

   b. Distributed systems should use some degree of data replication across nodes.

   c. In a distributed system, shared state is typically protected and synchronized across nodes by using semaphores.

   d. Nodes in a clustered distributed system are typically spread geographically.

   e. In a distributed system the nodes typically communicate and coordinate their actions by message passing.

   f. Linear horizontal scalability means that adding more nodes to a distributed system also adds total performance.

   g. Global distributed consensus is a typical example where scalability is obtained.

   h. When using hashing, the computational complexity and hashing time will spread in an unpredictable way.

   i. Marshaling (a k a serialization) provides an irreversible mapping to a standard format.

   j. Kademlia search for a target node is in the worst case done in O(log n) time, independently of the number of nodes in the system.

   k. The number of k buckets in Kademlia routing can be dynamically changed.

   l. Distributed Hash Tables (DHTs) can be completely decentralized.

   m. Imposing immutability of data complicates the design of distributed systems.

   n. DNS can be used for fail-over across web servers.

   o. In SQL databases there is a need to handle and resolve deadlocks.

   p. Blockchain solutions typically replicate all data to all nodes that are part.

   q. Leader based global consensus solutions typically assume that there can be a

small number of traitors.

r. NAT/firewall traversal by TURN is based on relaying all payload data through an external anchor point.

s. A pubsub service is typically implemented as a scalable distributed system that shards and replicates channels or topics to subsets of servers.

t. PGP requires Certificate Authorities (CA) to certify the ownership of public keys.

u. The pubsub paradigm is based on immutable messages that are consumed when read (i.e., read once).

v. In the Hadoop framework, a combiner function can always be used to decrease the data to be sent to the reducer.

w. Gaming servers and backends are often based on state updates by idempotent messaging.

x. For scalability reasons, online games tend to move heavy computation from the clients to the backends.

Note: In the following questions it is important to include sources (e.g., URLs) and that you explain the answer in your own words.

2. Miscellaneous (24p)

a) In distributed computing systems, what does it mean to be Byzantine? (2p)

b) What are the distinguishing features of microservices in terms of services-database mapping, event handling, multi-team software engineer workforce, and consistency of data? (4p)

c) In Bitcoin, explain the role and need of (at least) three different kind of hash functions that are used in the system. Note that a hash function that uses a key pair (public-private) is counted as one kind of hash function (not two). (6p)

d) In raft, what is replicated? Explain the need and use of so-called “terms”. (4p)

e) In Ethereum, can multiple smart contracts be executed concurrently? Explain. (6p)

f) Explain the typical approach to provide consistency in distributed datastores of mutable data/objects/files. (2p)

Best Wishes!
Finally, please fill out the online course evaluation. An email with a link will be sent to you tomorrow (Oct 27).